

Ghost Net — MacBook Air Task Sheet

Thursday 2026-09-03 | college day, laptop only, no GPU, no MARIDA, no credentials

Before anything

Every prompt below is self-contained — paste one, work it, then paste the next. They do not depend on each other and can be done in any order. All four are credential-free and need neither the GPU nor the MARIDA benchmark, so they genuinely run on the Air.

If the day fragments (likely): do Task 1 in a gap between lectures — it is short and self-closing. Save Task 2 for a contiguous block; it is the last unstarted PRD §12 acceptance bullet and the highest-value thing left. Task 4 is writing, so it survives interruption better than code.

Do not write OSCAR fetch or --oscar wiring. It cannot be tested without EARTHDATA_TOKEN, and untested code on a headline number's path is the one thing this project has repeatedly been burned by.

Task 1 — Fold the generalisation result into the ablation study

Time ~30 min

Value Closes a stale framing in graded material

docs/ablation-study.md §4 presents region recall 0.703 with no indication it is a *within-tile* number. That is now measured and the doc should say so.

Continue the Ghost Net project. Read MACHINE-WORKFLOW.md Status Log, top entries. This is the MacBook Air (no GPU); verify with python scripts/check_machine.py. PULL FIRST.

Task: fold the geographic-generalisation result into docs/ablation-study.md section 4, which is your file.

Section 4 currently presents region recall 0.407 -> 0.703 with no indication that it is a WITHIN-TILE number. It is: MARIDA splits by patch, not by tile, so 327 of 359 test patches (91%) sit on MGRS tiles the model trained on. The comparison is still fair (the FDI is scored through the identical path) but it is not evidence of generalisation, and section 4 currently reads as though it were.

The measurement now exists. From eval/results.md, section "Geographic generalisation", over the IDENTICAL 84 18QYF patches:

detector_v1 (trained on 18QYF)	P=0.9355	R=0.9254	F1=0.9304
holdout (never saw 18QYF)	P=0.9085	R=0.8129	F1=0.8581
cost of the region being unseen	-0.0270	-0.1125	-0.0723

About 7 F1 points lost on unseen water, almost all of it RECALL; precision barely moves. It finds less on new water but what it flags stays trustworthy.

Artefacts: eval/holdout_18QYF.json (the holdout model) and eval/holdout_18QYF_leaky.json (detector_v1 on the same patches). Cross-check the numbers against those rather than trusting this prompt.

TWO THINGS THAT MUST SURVIVE THE EDIT:

1. Do NOT quote the holdout score minus the in-distribution test score as a gap. 18QYF carries 13.24 debris px/patch against 0.98 for the rest of test, so that subtraction measures task difficulty, not distribution shift, and comes out the wrong sign (-0.194). Only the paired table above is valid. eval/results.md explains this under "One number in this experiment must not be quoted".
2. Carry the five caveats across, or point at them: one region, one seed, no repeats; the holdout model trained on 8.5% less data so -0.072 is an UPPER bound; Haiti is the same current system as the demo region, so this is an unseen TILE in the western Caribbean, not a different ocean.

Then update the Status Log and push.

Task 2 — Evidence traceability (PRD §12, last unstarted bullet)

Time 2-3 h, wants a contiguous

block

Value Highest — graded acceptance criterion

\$12's final bullet: *"Every output in the demo can be traced back to its underlying evidence on request."*

Nothing implements or demonstrates it. Your own ablation write-up did this by hand against eval/*.json and found two real defects — which is the argument for making it a check that cannot rot.

```
Continue the Ghost Net project. Read MACHINE-WORKFLOW.md Status Log, top
entries. This is the MacBook Air (no GPU); verify with
python scripts/check_machine.py. PULL FIRST.
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```
Task: PRD section 12's last acceptance bullet -- "Every output in the demo can
be traced back to its underlying evidence on request (imagery tile, current
field, vessel record)". It is the only section 12 bullet with nothing built or
demonstrated against it.
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Build it in two halves, and do the first one first.
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HALF A -- A REPEATABLE RECONCILIATION CHECK (this is the real deliverable).
When you wrote docs/ablation-study.md you cross-checked every quoted figure
against eval/*.json by hand and found two genuine defects: a mis-transcribed
CNN candidate table, and three FR-2.2 arms with no committed artefact. Both are
now fixed. Doing that by hand once is how they were caught; doing it never
again is how the next one ships.
```

```
So: a test or script that asserts every number the console serves reconciles
with the eval/*.json file it claims to come from. src/ghostnet/benchmark.py is
the right seam -- it already reads, reshapes and attaches caveats rather than
computing, so the check is "does what the endpoint serves still equal what the
JSON says". Make a mismatch FAIL, with a message naming the field, both values
and the source file.
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Artefacts currently in eval/: marida_ablation, detector_fdi_test,
detector_cnn_test, multitemporal, multitemporal_18QYF_2020-03,
multitemporal_18QYF_2020-11, multitemporal_16PCC_2018-09,
multitemporal_sensitivity_010, holdout_18QYF, holdout_18QYF_leaky.
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Note the last four are NOT read by benchmark.py today. Deciding whether they
should be surfaced is part of this task -- but a number that no artefact backs,
or an artefact that nothing checks, is exactly what this bullet exists to stop.
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HALF B -- THE WALK-THROUGH. Then make the trace demonstrable end to end: from a
detection in the console, to its evidence refs, to the imagery tile ID, current
field and vessel record behind it. The export schema already carries evidence
refs and provenance (thresholds actually used, tile IDs, ensemble/seed, git
commit), so this is surfacing and documenting a path that mostly exists, not
inventing one. Write it up as the sequence an evaluator would actually be
walked through in the viva.
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CONSTRAINTS:
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- Tests use synthetic fixtures only: no network, no credentials, no downloaded data, so the suite still runs on a fresh clone. The reconciliation check reads committed eval/*.json, which is fine -- those are committed artefacts, not downloaded data.
- The artefact on the console is still SYNTHETIC and must keep saying so. A real run is blocked on datasets nobody has downloaded.
- Do not compute a metric a second way. eval/results.md is the authority; the check confirms agreement, it does not re-derive.

```
Then update the Status Log and push.
```

Task 3 — Put the generalisation number on the console

Time 1-2 h

Value Closes an honesty gap that is documented but not visible

Two new artefacts exist that nothing reads, and the metrics strip still shows 0.703 with no sign it is within-tile. This is the same shape as the FR-2.2 work you already did — a warning-toned metric with its caveat on the face.

Continue the Ghost Net project. Read MACHINE-WORKFLOW.md Status Log, top entries. This is the MacBook Air (no GPU); verify with python scripts/check_machine.py. PULL FIRST.

Task: surface the geographic-generalisation result on the operator console.

Two artefacts landed that nothing reads: eval/holdout_18QYF.json and eval/holdout_18QYF_leaky.json. Meanwhile the metrics strip shows region recall 0.703 with nothing saying it is a WITHIN-TILE number -- 327 of 359 MARIDA test patches (91%) sit on tiles the model trained on, because MARIDA splits by patch rather than by tile.

This is the same shape as the FR-2.2 work you already did: a measured caveat that an evaluator would otherwise not know to ask about, put on the face of the app rather than left in eval/results.md.

The number, over the identical 84 18QYF patches:

detector_v1 (trained on 18QYF)	F1=0.9304	
holdout (never saw 18QYF)	F1=0.8581	-> cost -0.0723, mostly recall

SUGGESTED SHAPE, but use your judgement -- you own this UI:

- benchmark.py reads the two holdout artefacts the way it already reads the detector ones. It reads and reshapes; it does not recompute.
- The strip gains a fourth item beside region recall, warning-toned, saying region recall 0.703 is within-tile and naming the -0.072 unseen-region cost.
- Expanding gives the paired table, the 84-patch sample size, and the caveats.

THE TRAP THIS MUST NOT FALL INTO, and it is the reason to be careful: do NOT show the holdout model's in-distribution test score (0.6637) next to its 18QYF score (0.8581) as a before/after. 18QYF carries 13.24 debris px/patch against 0.98 for the rest of test, so that pairing measures task difficulty and reads backwards -- it makes the unseen region look EASIER. Only the paired table over identical patches is meaningful. eval/results.md says this in full under "One number in this experiment must not be quoted", and scripts/train_cnn.py prints a warning about it. Consider a test that the endpoint refuses to serve that pairing, in the spirit of test_benchmark_never_serves_the_gain_without_the_region_recall.

Also: the generalisation figures come from a checkpoint that is NOT the pipeline's detector. detector_v1.pt is unchanged and still the detector; detector_holdout_18QYF.pt is an experiment artefact and is not committed. The UI must not imply the shipped detector is the holdout one.

Then update the Status Log and push.

Task 4 — Report and viva materials

Time open-ended,
interruption-tolerant

Value Graded, unstarted

Good filler for fragmented time. docs/ablation-study.md is the model: it assembles from committed artefacts and states what it cannot verify.

Continue the Ghost Net project. Read MACHINE-WORKFLOW.md Status Log, top entries, and docs/ablation-study.md. This is the MacBook Air (no GPU); verify with python scripts/check_machine.py. PULL FIRST.

Task: report and viva materials. Graded, unstated, and it needs no credentials.

docs/ablation-study.md is the model to follow: assembled by reading committed eval/*.json rather than by re-running anything, and explicit about what it could not verify.

What the project can actually claim, all measured and all reproducible from a committed artefact:

- FR-2.4 verification ablation: precision 0.238 -> 0.623, F1 0.385 -> 0.753 on real MARIDA, thresholds fitted on train only, baseline restored from literature values so the "before" arm is genuinely reproducible.
- FR-1.4 CNN detector: region recall 0.407 -> 0.703, 7.6x fewer candidates.
- The subsumption finding: verification's contribution falls from +0.385 over the FDI to +0.080 over the CNN. Quote BOTH; never +0.385 alone once the CNN is the detector.
- Geographic generalisation: -0.072 debris F1 on an unseen region, almost all recall. Region recall 0.703 is a within-tile number.
- FR-2.2 multi-temporal: a measured NEGATIVE. It contributes nothing today and is blocked on FR-3.1's missing current field. Report as a dependency, never as a contribution. PRD section 12 records it as a measured exception to the every-agent-is-load-bearing test -- keep that framing.

What must NOT be claimed:

- Drift validation. No drifter passed through the demo region during the demo window -- zero observations. If it is ever run it is a model check over other years, never validation of the demo run.
- Any end-to-end run on real data. Still blocked on four datasets.
- Anything from run_pipeline_demo.py -- synthetic, illustrative, not results.

The governing rule: no number reaches the report that you cannot derive and defend. An evaluator asking "why is this 0.703?" is asking the person.

Then update the Status Log and push.

What is still blocked, and why none of it is Air work

FR-2.2's re-measurement, FR-3 drift validation, FR-4 attribution, FR-5 dark-vessel correlation and FR-6.1 ecological risk all wait on the same four things — two free signups and two downloads, none of which need a GPU and none of which have been done:

Needs	How	Unblocks
EARTHDATA_TOKEN	urs.earthdata.nasa.gov -> Generate Token -> .env	OSCAR, FR-2.2 re-run, FR-3
GFW_API_TOKEN	globalfishingwatch.org/our-apis -> .env	FR-5
Protected Planet	WDPA marine subset, then build_region_extracts.py mpa	FR-6.1
River table	Meijer 2021 global CSV, then build_region_extracts.py rivers	FR-4

These are account and download work, so they are yours on whichever machine — but the two extract commands need the downloaded source files, and the workstation is where the pipeline gets exercised. Creating the two tokens is the single highest-value thing available, and it takes about ten minutes.